* Digital Communications *

SALEH RAMADAN

" Computer "

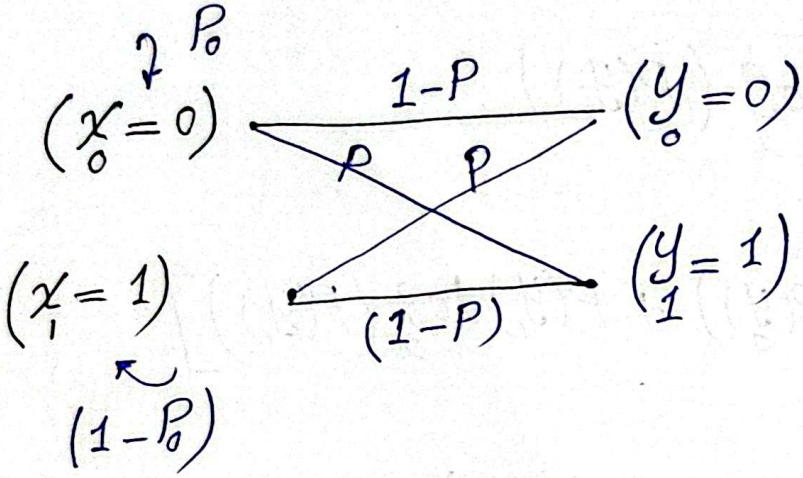
* Sheet 1 "

"Information Theory"

$$H(z) = z \log_2\left(\frac{1}{z}\right) + (1-z) \log_2\left(\frac{1}{1-z}\right)$$

$$z = P_0 P + (1-P_0)(1-P)$$

*** Problem 1:**



Solution

a) $I(x,Y) = H(x) - H(x|Y)$

$= H(Y) - H(Y|x) \Longrightarrow$ It's better To use This as get P(y) is easy Than Get P(x|Y)

$$P(y_0) = P(y_0|x_0)\,P(x_0) + P(y_0|x_1)P(x_1)$$

$$P(y_0|x_0) = 1-P \;,\; P(y_0|x_1) = P \;,\; P(x_0) = P_0 \;,\; P(x_1) = 1-P_0$$

CamScanner

②

$$P(y_0) = (1-P)P_0 + P*(1-P_0)$$

$$P(y_1) = P(y_1|x_0)\,P(x_0) + P(y_1|x_1)P(x_1)$$

$$P(y_1|x_0) = P \quad , \quad P(y_1|x_1) = 1-P$$

$$P(y_1) = P*P_0 + (1-P)(1-P_0)$$

$$H(y) = \sum_i - P(y_i)\log_2\left(P(y_i)\right)$$

$$= -\left[P(y_0)\log_2(P(y_0)) + P(y_1)\log_2(P(y_1))\right]$$

$$H(y) = -\Big[\left[(1-P)P_0 + P(1-P_0)\right]\log_2(P(y_0)) + PP_0 + (1-P)(1-P_0)\log_2(P(y_1))\Big]$$

$$P(y_1) = z, \quad P(y_0) = 1-z$$

$$H(y) = P(y_0)\log_2\left(\frac{1}{P(y_0)}\right) + P(y_1)\log_2\left(\frac{1}{P(y_1)}\right)$$

$$H(y) = (1-z)\log_2\left(\frac{1}{1-z}\right) + z\log_2\left(\frac{1}{z}\right)$$

CamScanner

$$H(y) = JH(z) \longrightarrow (1)$$

$$H(y|x) = \sum_x \sum_y P(x)\, P(y|x) \log_2 \left(\frac{1}{P(y|x)}\right)$$

$$= P(x_0) P(y_0|x_0) \log_2 \left(\frac{1}{P(y_0|x_0)}\right)$$

$$+ P(x_1) P(y_0|x_1) \log_2 \left(\frac{1}{P(y_0|x_1)}\right)$$

$$+ P(x_0) P(y_1|x_0) \log_2 \left(\frac{1}{P(y_1|x_0)}\right)$$

$$+ P(x_1) P(y_1|x_1) \log_2 \left(\frac{1}{P(y_1|x_1)}\right)$$

$$H(y|x) = P_0 (1-P) \log_2 \left(\frac{1}{1-P}\right)$$

$$+ (1-P_0) P \log_2 \left(\frac{1}{P}\right)$$

$$+ P_0 P \log_2 \left(\frac{1}{P}\right)$$

$$+ (1-P_0)(1-P) \log_2 \left(\frac{1}{1-P}\right)$$

$$H(y|x) = \log_2 \left(\frac{1}{1-P}\right) \left[P_0 - P_0 P + 1 - P - P_0 + P P_0\right]$$

$$+ \log_2 \left(\frac{1}{P}\right) \left[P(1 - P_0 + P_0)\right]$$

CamScanner

4)

$$H(y|x) = P \log_2\left(\frac{1}{P}\right) + (1-P)\log_2\left(\frac{1}{1-P}\right)$$

$$\boxed{H(y|x) = \mathcal{H}(P)} \longrightarrow (2)$$

From (1) & (2): $\boxed{I(x,y) = \mathcal{H}(z) - \mathcal{H}(P)}$ ✳✳

* Remember: $\log_2(x) = \frac{\ln(x)}{\ln(2)}$

b) $\frac{dI(x,y)}{dP_0} = 0$

$$\frac{d\mathcal{H}(z)}{dP_0} = 0 \Longrightarrow \frac{d\mathcal{H}(z)}{dz} = 0$$

$z \to f(P_0)$

$$\mathcal{H}(z) = z\log_2\left(\frac{1}{z}\right) + (1-z)\log_2\left(\frac{1}{1-z}\right)$$

$$\frac{dH(z)}{dP_0} = -\left[z\left[\frac{1}{\ln(2)}\cdot\frac{1}{z} * z'\right] + z'\log_2(z)\right.$$

$$\left. + \frac{(1-z)}{\ln(2)} * \frac{-z'}{(1-z)} + \log_2(1-z) * -z'\right]$$

$$\frac{dH(z)}{dP_0} = 0$$

5)

$$\frac{dH(z)}{dP_0} = \frac{-z'}{\ln(2)} - z' \log_2(z) + \frac{z'}{\ln(2)} + z' \log_2(1-z)$$

$$= -z' \left[\log_2(z) - \log_2(1-z) \right]$$

$$\log_2(z) = \log_2(1-z)$$

$$z = 1 - z \implies z = \frac{1}{2}$$

$$z = P_0 P + (1-P_0)(1-P)$$

$$P_0 P + 1 - P - P_0 + P_0 P = \frac{1}{2}$$

$$2 P_0 P - P - P_0 = \frac{-1}{2}$$

$$P_0 - 2 P_0 P = \frac{1}{2} - P$$

$$P_0 (1 - 2P) = \frac{(1-2P)}{2} \implies \boxed{P_0 = \frac{1}{2}}$$

c) $C = I(X,Y)_{@ P_0 = \frac{1}{2}}$, $I(X,Y) = H(z) - H(P)$ $\rightsquigarrow z = \frac{1}{2}$

$$H(z) = \frac{1}{2} * \log_2(2) + \frac{1}{2} \log_2(2) = 1$$

$$\implies \boxed{I = 1 - H(P)}$$

* Problem 2:

$$P(y|x) = \begin{array}{c} \\ x_0 \\ x_1 \end{array} \begin{array}{c} \begin{array}{cc} y_0 & y_1 \end{array} \\ \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{10} & \frac{9}{10} \end{bmatrix} \end{array}$$

$$P(x=0) = \frac{1}{3} \quad , \quad P(x=1) = \frac{2}{3}$$

Solution

a) $$H(x) = \sum_x -P(x) \log_2 (P(x))$$

$$= -\left[P(x_0) \log_2 (P(x_0)) + P(x_1) \log_2 (P(x_1))\right]$$

$$H(x) = 0.918 \text{ bits} \quad \#$$

$$H(y) = \sum_y -P(y) \log_2 (P(y))$$

$$P(y_0) = P(y_0|x_0) P(x_0) + P(y_0|x_1) P(x_1)$$

$$P(y_1) = P(y_1|x_0) P(x_0) + P(y_1|x_1) P(x_1) = 1 - P(y_0)$$

$$P(y_0) = \frac{13}{45} \quad , \quad P(y_1) = \frac{32}{45}$$

CamScanner

$$H(y) = -\left[P(y_0) \log_2 (P(y_0)) + P(y_1) \log_2 (P(y_1)) \right]$$

$$\boxed{H(y) = 0.867 \text{ bits.}}$$

$$H(y|x) = \sum_x \sum_y -P(x)\, P(y|x) \log_2 (P(y|x))$$

$$= -\left[P(x_0) P(y_0|x_0) \log_2 (P(y_0|x_0)) \right.$$
$$+ P(x_1) P(y_0|x_1) \log_2 (P(y_0|x_1))$$
$$+ P(x_0) P(y_1|x_0) \log_2 (P(y_1|x_0))$$
$$\left. + P(x_1) P(y_1|x_1) \log_2 (P(y_1|x_1)) \right]$$

$$\boxed{H(y|x) = 0.6187 \text{ bits.}}$$

---

b) $I(x,y) = H(x) - H(x|y)$

$= H(y) - H(y|x)$

$$\boxed{I(x,y) = 0.2483 \text{ bits}}$$

$$\boxed{H(x|y) = 0.6697 \text{ bits.}}$$

CamScanner